

Lock-in Progress Handoff

What is working now

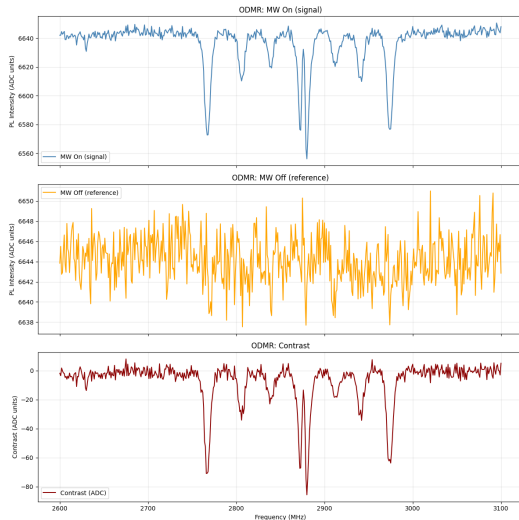
- ▶ The notebook path is ODMR sweep, parked-frequency plan, multipoint lock-in acquire, two-point peak inference, then vector reconstruction.
- ▶ The parked-frequency acquire now measures 16 nonuniform frequencies in one uploaded FPGA program through `MultipointLockinODMR`.
- ▶ Reference readouts use an MW off data by setting a delay time of MW on/off at around 1700 μ s.
- ▶ Live mode builds one program outside the loop and asserts `pre_init=False`; this prevents the recurring per-batch peaking transient.

Notebook run snapshot

ODMR source	20260507_140754
Parked points	16
Transitions	8
Bias field	-2.30, -2.99, -3.14 mT
Single-batch acquire	116 ms
Live run	281 samples, 21.0 s
Live rate	13.4 Hz
Vector rank-ok	281 / 281

Sources read only: `Lockin_module.ipynb` and
`LOCKIN_ACQUIRE.md`.

MW-Off Reference Issue Resolved



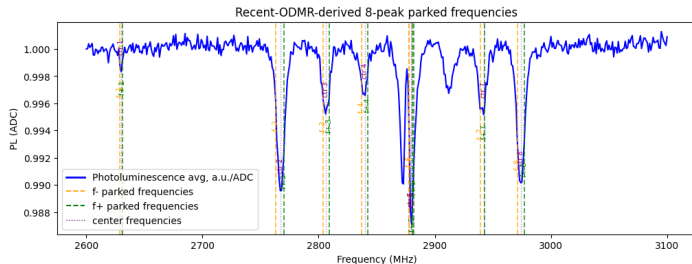
What changed

- ▶ The ODMR cell now records both MW-on signal and MW-off reference in the same sweep.
- ▶ MW-off is separated using a long MW on/off delay, about 1700 us, so the reference no longer carries the ODMR dips.
- ▶ The contrast trace is now the clean signal-minus-reference response used for planning parked frequencies.

Why new staff should care

All downstream lock-in math assumes each parked point has a paired reference count. If the reference still contains resonance structure, the normalization and two-point frequency inference are biased.

Parked-Frequency Plan from ODMR



Plan table written by notebook

Blk	Center MHz	Line MHz	f_- MHz	f_+ MHz
1	2630.0	4.0	2629.0	2632.0
2	2767.4	9.9	2764.0	2770.0
3	2806.0	4.0	2803.0	2809.0
4	2840.0	4.0	2837.0	2843.0
5	2880.6	3.4	2878.0	2882.0
6	2880.6	3.6	2878.0	2882.0
7	2941.5	4.5	2939.0	2944.0
8	2974.9	10.5	2971.0	2978.0

Why this matters

The custom program is needed because these parked points are not uniformly spaced, so a normal linear Qick sweep is not the right abstraction.

Single-Program Multipoint Acquire

Notebook output

- ▶ Built one MultipointLockinODMR program for 16 frequencies.
- ▶ Predicted FPGA work: 68 ms per batch, or 6.8 ms per rep times 10 reps.
- ▶ Actual first batch in the saved notebook output: 116 ms.
- ▶ Output: `multipoint_lockin_collected.csv`, one wide row with `peak_NN`, `peak_NN_ref`, and `peak_NN_freq_mhz`.

Guardrail for new staff

Do not instantiate `MultipointLockinODMR` inside a batch loop. Build once, then loop over `prog.acquire(progress=False)`.

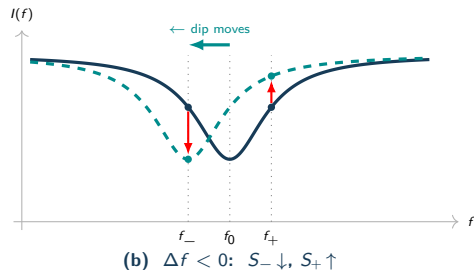
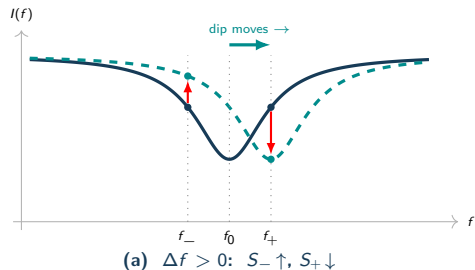
Example acquired columns

Peak	Freq MHz	Signal ADC	Ref ADC
01	2629.0	6529.6	6535.3
02	2632.0	6539.4	6543.4
09	2878.0	6502.5	6507.1
10	2882.0	6488.6	6498.9
15	2971.0	6499.1	6509.3
16	2978.0	6499.2	6510.1

Timing comparison in docs

Old 17-program loop at `reps=100`: about 6.6 s per batch in a good case, worse with network jitter. New multipoint path: about 700 ms per batch at `reps=100` after the one-time upload.

Math 1/2: Dip Shift Causes Asymmetric Intensity Change



Why park on the flanks instead of re-sweeping

The reference ODMR sweep tells us where each dip sits. We park at f_- and f_+ on the steep flanks (one MW frequency on each side of $f_{0,k}$). When the dip shifts by Δf , the two parked intensities respond in *opposite* directions: one flank exits the dip (intensity rises), the other gets pulled deeper into it (intensity falls). The signed difference $S_+(t) - S_-(t)$ encodes both the *sign* and the *size* of Δf in a single scalar — no re-sweeping needed.

Slope is the conversion factor

The rate at which $S_+ - S_-$ changes per MHz of dip shift is the slope difference $m_- - m_+$ (units: intensity / MHz). It is a property of the dip's *shape*, captured once during the reference ODMR sweep. The same intensity drop on a sharp dip vs. a broad dip implies very different Δf , so without the slope you cannot turn an intensity error into a frequency error. Slide 2 shows the formal derivation.

Math 2/2: Linearization, Δf Formula, and Example

Linearize the spectrum at the parked points

The dip shape depends only on $(f - f_0)$. A small shift $f_0 \rightarrow f_0 + \Delta f$ changes the intensity at fixed f to first order by

$$I(f, f_0 + \Delta f) \approx I(f, f_0) - \left. \frac{\partial I}{\partial f} \right|_{f_0} \Delta f.$$

At the two parked points this gives

$$S_-(t) \approx b_- - m_- \Delta f, \quad S_+(t) \approx b_+ - m_+ \Delta f,$$

and subtracting cancels common-mode drift in $b_{\pm}$:

$$\underbrace{(S_+ - S_-)_{\text{now}}}_{\Delta d_{\text{now}}} - \underbrace{(b_+ - b_-)}_{\Delta d_{\text{ref}}} = (m_- - m_+) \Delta f.$$

Solve for Δf :

$$\Delta f_k = \frac{\Delta d_{\text{now}} - \Delta d_{\text{ref}}}{m_- - m_+}, \quad f_{\text{new},k} = f_{0,k} + \Delta f_k$$

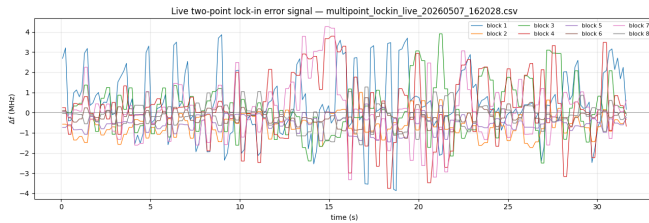
Symbol legend and sanity checks

- ▶ $b_{\pm}$: normalized PL at $f_{\pm}$ from the reference ODMR sweep.
- ▶ $m_{\pm} = dI/df$ at $f_{\pm}$. Notebook asserts $m_- < 0 < m_+$ (left flank descends, right flank ascends), so the denominator $m_- - m_+ < 0$.
- ▶ $\Delta d_{\text{now}} = S_+(t) - S_-(t)$ from the latest batch; $\Delta d_{\text{ref}} = b_+ - b_-$ from the calibration sweep.
- ▶ Sign: positive Δf from panel (a) gives $\Delta d_{\text{now}} < \Delta d_{\text{ref}}$, i.e. negative numerator. Negative $\div$ negative = positive Δf . ✓
- ▶ Symmetric-flank limit ($m_- = -m_+$) recovers the textbook $\Delta f = (\Delta d_{\text{now}} - \Delta d_{\text{ref}})/(2m_-)$.
- ▶ Field projection along this transition's NV axis: $\Delta B_{\text{proj},k} = \Delta f_k / \|df/dB\|_k$.

Latest batch, block 4 (numbers from notebook)

m_-, m_+	$-4.17 \times 10^{-4}, +5.36 \times 10^{-4}$ per MHz
Δd_{now}	-1.47×10^{-4}
Δd_{ref}	$+7.96 \times 10^{-4}$
$\Delta d_{\text{now}} - \Delta d_{\text{ref}}$	-9.43×10^{-4}
Δf	$+0.990$ MHz
f_{new}	2862.979 MHz

Live Lock-in Run: Drift and Noise Floor



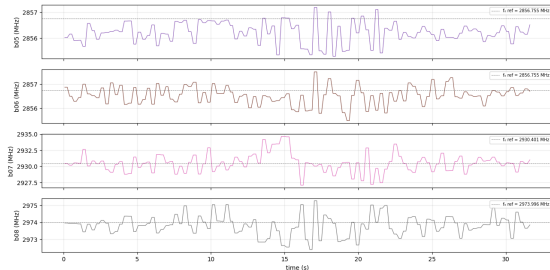
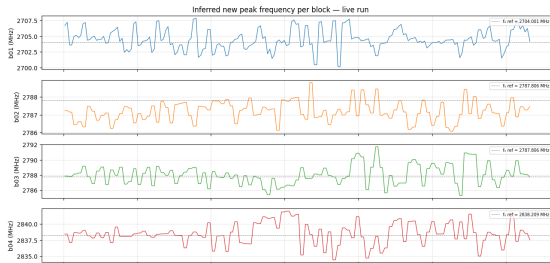
Live run summary

- ▶ Run: live CSV 20260507_141355.
- ▶ 281 batches over 21.0 s, 13.4 Hz update rate.
- ▶ Timing line: predicted 68 ms; actual median 12 ms, p95 15 ms, max 17 ms.
- ▶ Watchdog never fired in the saved run.

Per-block live statistics

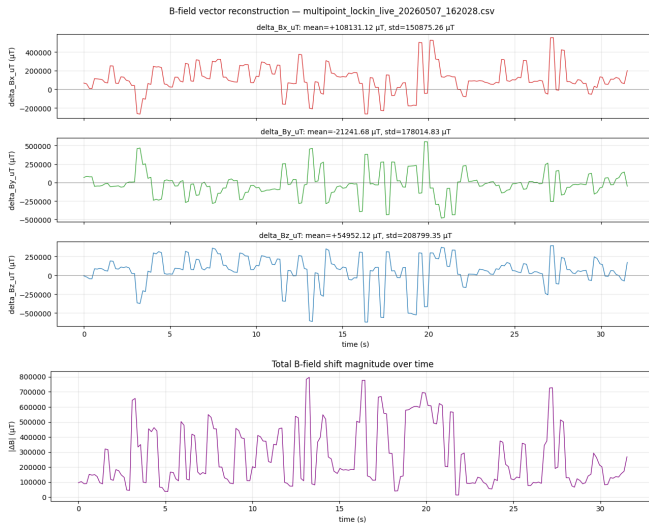
Blk	mean Δf MHz	std Δf MHz	std ΔB uT
1	-0.211	0.354	12.6
2	-0.062	0.172	6.1
3	-0.583	0.460	16.3
4	+0.973	0.619	22.3
5	-0.645	0.102	7.4
6	-0.723	0.099	7.2
7	+0.024	0.279	9.9
8	-0.201	0.128	4.6

Live Inferred New Peak Frequencies



Notebook post-processing figure: each block plots $f_{new}(t) = f_0 + \Delta f(t)$ with the dashed reference f_0 from the ODMR calibration.

B-field Reconstruction Output



Notebook reconstruction status

- ▶ Input: wide live CSV from the same 281-sample run.
- ▶ Status: ok.
- ▶ Samples: 281.
- ▶ Rank-ok: 281.
- ▶ Matrix condition number: 14.08.

Vector component statistics

Comp.	Mean uT	Std uT	Range uT
ΔB_x	18,923	81,622	-351,633 to 248,403
ΔB_y	-290,071	221,862	-942,851 to 688,841
ΔB_z	79,653	104,794	-323,599 to 634,124

Immediate next checks

Before handing the system to new operators, replot peak_01 and acq_seconds from CSV outside the live cell, then verify no recurring batch-edge spike is present.